

RAJGURU MATHIYALAGAN

Data Scientist | Machine Learning | Python | SQL | Azure ML

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PROFESSIONAL SUMMARY

Final-year B.Sc. IT student with hands-on experience in **machine learning**, **statistical modeling**, and **data analysis**. Built end-to-end **data pipelines** processing 1.6M+ records using Python, SQL, and **Azure Machine Learning Studio**. Skilled in **feature engineering**, **model evaluation**, ETL workflows, and translating **business intelligence** into **stakeholder-ready** dashboards and actionable insights.

TECHNICAL SKILLS

Languages: Python, SQL

Libraries: Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Flask

ML Techniques: Supervised/Unsupervised Learning, Ensemble Learning, Gradient Boosting, Random Forest, Classification, Regression, K-Means, SMOTE, Cross-Validation, Hyperparameter Tuning

Data Skills: EDA, Data Cleaning, Feature Engineering, Statistical Modeling, Statistical Analysis, A/B Testing, ETL, Model Evaluation, Data Visualization

Cloud & ML Platforms: Microsoft Azure, Azure Machine Learning Studio, Azure AutoML, Azure ML Designer

Tools: Jupyter, Git/GitHub, Tableau, Power BI, Google Colab, VS Code, Microsoft Excel

Deployment: Flask, React.js, REST API, PostgreSQL

PROJECTS

PropVision | UAV Performance Prediction Platform

[GitHub](#) | [Live Demo](#)

End-to-End Data Pipeline | Cloud ML | Azure AutoML | Full-Stack Deployment

- Built an end-to-end **data pipeline** and ML model predicting UAV propeller thrust, power, and efficiency across 1.6M records, achieving 90% accuracy with **XGBoost** and **Gradient Boosting** via **Azure Machine Learning Studio**.
- Implemented **Azure AutoML** regression workflows and **Azure ML Designer** pipelines for cloud-based model training, **hyperparameter tuning**, and **cross-validation**.
- Applied physics-informed **feature engineering** (Blade Area, Solidity) via numerical integration to improve model generalization.
- Deployed full-stack solution with React.js frontend, Flask **REST API**, PostgreSQL database, and **Tableau dashboarding** to surface **business intelligence** for **cross-functional stakeholder** reporting.

Employee Turnover Analysis | HR Attrition Prediction

[GitHub](#)

HR Analytics | Classification | Statistical Modeling

- Built **statistical modeling** and **classification pipeline** achieving 90% accuracy and 0.90 ROC-AUC using **Random Forest** and **Logistic Regression** with **cross-validation**.
- Designed ETL workflow including SMOTE oversampling and data wrangling to address class imbalance and improve minority class prediction reliability.
- Conducted EDA, **K-Means Clustering**, and **A/B testing** of retention strategies to identify key attrition drivers; delivered **business intelligence dashboarding** and **stakeholder-ready** insights to support **cross-functional** HR decision-making.

EDUCATION

B.Sc. Information Technology — D.G. Ruparel College, Mumbai University

2023 – 2026

CGPA: 8.60 / 10 | **Coursework:** Statistics, Data Structures, DBMS, Linear Algebra, Operating Systems

CERTIFICATIONS

Introduction to Programming Using Python — Simplilearn

2024

Python fundamentals, OOP, file handling, and data libraries (NumPy, Pandas) for data-driven applications.

Data Science with Python — Simplilearn

2025

Data wrangling, ETL, EDA, statistical modeling, supervised & unsupervised learning, A/B testing, model evaluation, and business intelligence workflows.

Data Analytics Job Simulation — Deloitte (Forage)

2026

Practical tasks in data analysis and forensic technology, including spreadsheet-based analysis and data classification.